ASSISTANT PROFESSOR

Advancing Intelligent Control and Autonomous Robotic Systems

Research at the intersection of control theory, learning, and robotics, with emphasis on safe, efficient, and experimentally validated autonomous systems.

Explore Research → | View Publications →

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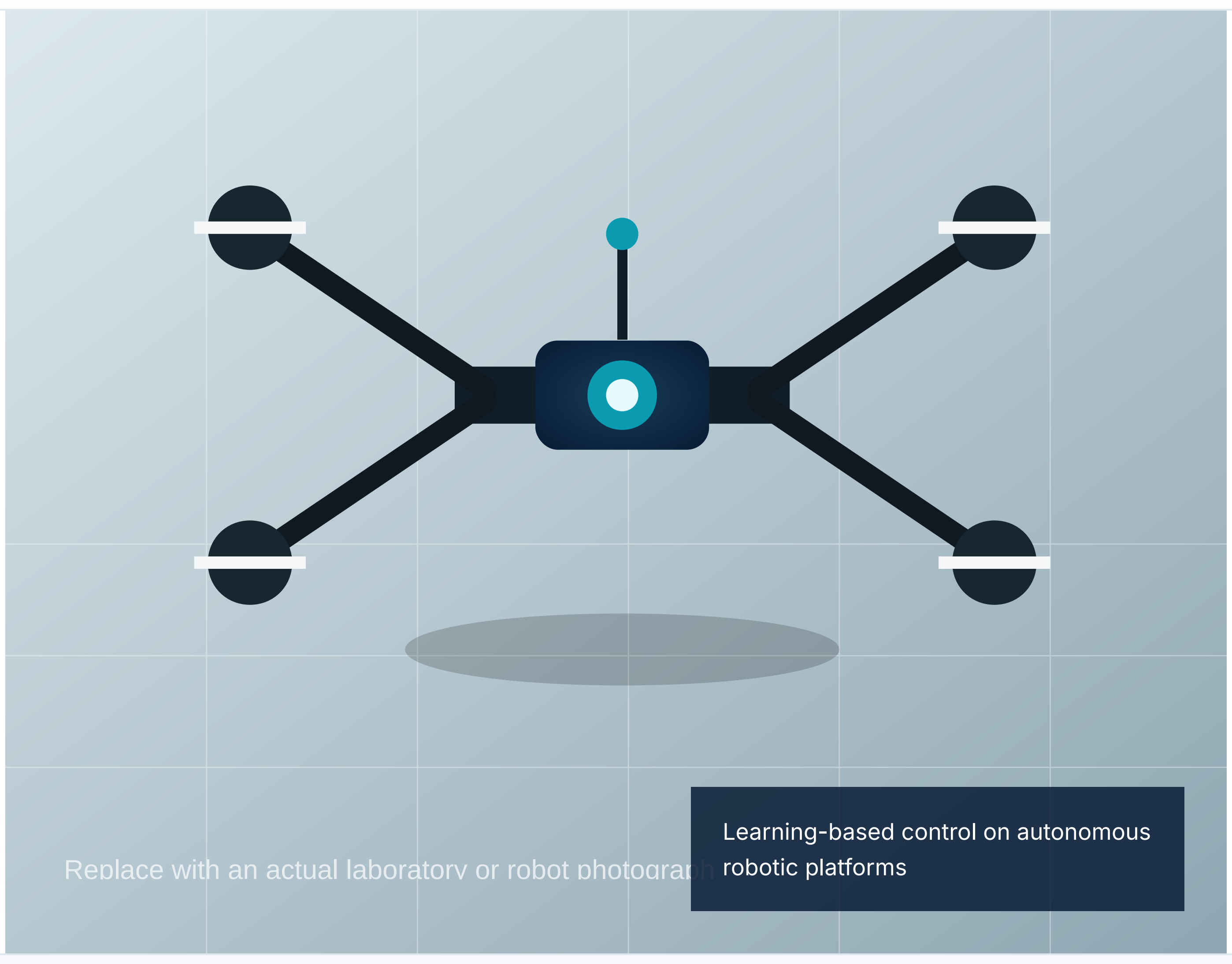
LOCATION
Sharjah, UAE



Learning-based control on autonomous robotic platforms

01 RESEARCH AREAS

View all research areas →

01
ced Control
Adaptive, robust, and optimal control methods for nonlinear and uncertain systems
Learn more →

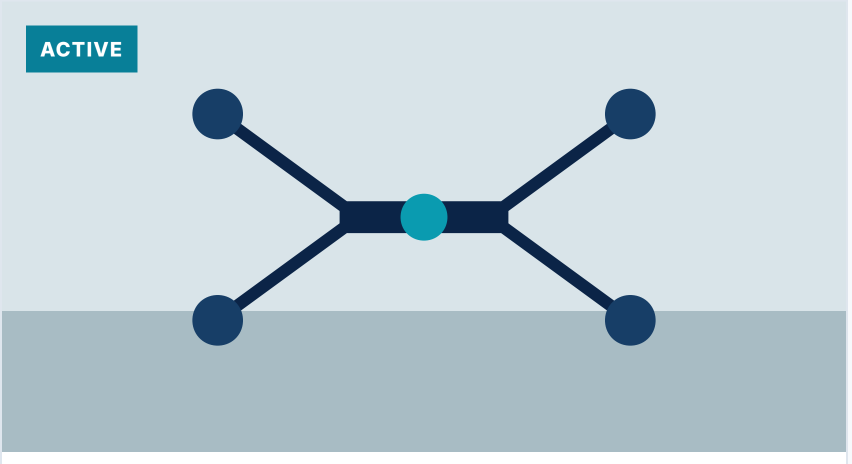
02
ng-Based Control
Reinforcement learning and data-driven methods for control, adaptation, and decision making
Learn more →

03
mous Robotics
Perception, planning, estimation, and control for autonomous robotic platforms.
Learn more →

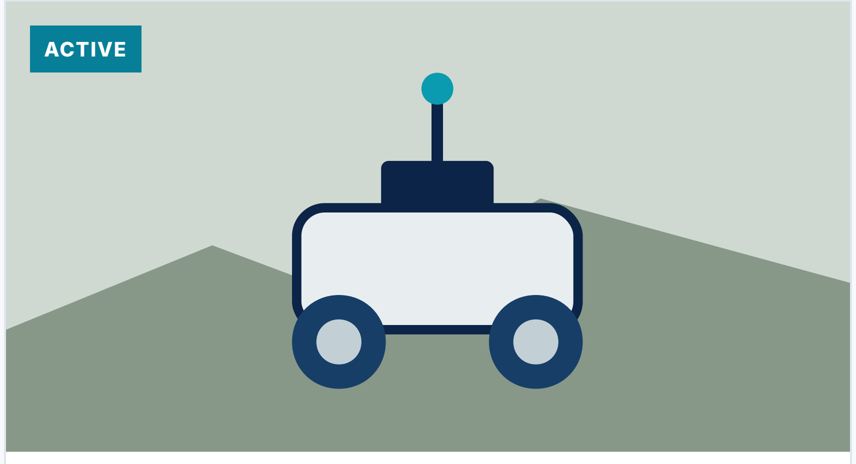
04
ent Systems
Data-driven modeling, system identification, and intelligent optimization.
Learn more →

02 FEATURED PROJECTS

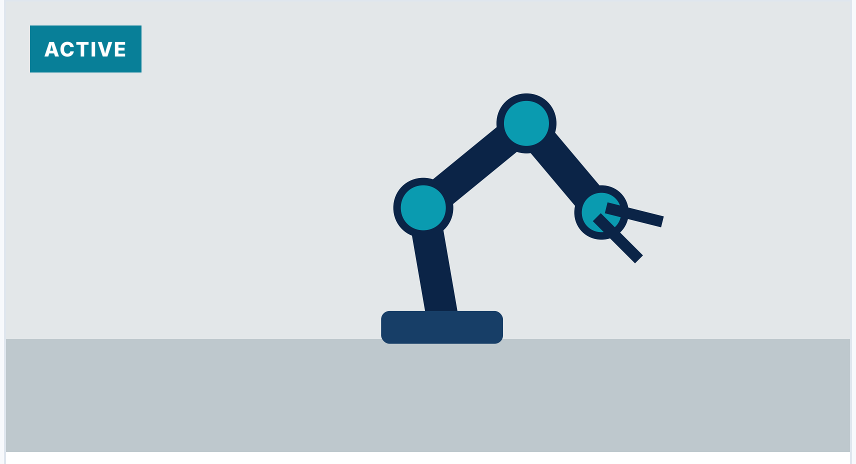
View all projects →



Learning-Augmented Adaptive Control
Integrating model-based adaptive control with reinforcement learning for enhanced performance under uncertainty and constraints.
Adaptive Control · RL · Quanser



Autonomous Navigation & Control
Developing perception, planning, and control algorithms for safe and efficient navigation on autonomous mobile platforms.
ROS 2 · Navigation · MPC



Learning for Robotic Manipulation
Learning-based control and perception for dexterous and adaptive robotic manipulation tasks.
Learning · Vision · Manipulation

4	3	4	4
Publications	Active Projects	Graduate Researchers	Research Areas

SELECTED PUBLICATIONS

View all →

1. **Learning-Augmented Adaptive Control for Nonlinear Systems with Safety Guarantees**
IEEE Transactions on Automatic Control, 2026
Abstract PDF Code
2. **Safe Reinforcement Learning for Constrained Robotic Control**
IEEE ICRA, 2025
Abstract Video
3. **Robust Terrain-Aware Navigation Using Predictive Control**
IEEE RA-L, 2025
Abstract
4. **Data-Driven System Identification with Stability Guarantees**
International Journal of Control, 2024
Abstract

LATEST NEWS

View all →

AUG 20, 2026
Our paper on learning-augmented adaptive control was accepted for publication.

JUL 15, 2026
A new experimental project on robust autonomous navigation has started.

JUN 03, 2026
A new graduate researcher joined the group to work on learning-based robotic control.

MAY 08, 2026
The team completed a closed-loop navigation demonstration on the mobile robot platform.

APR 19, 2026
Presented recent work on safe learning-based control for autonomous systems.

CURRENT RESEARCHERS

Meet the group →

AK
Alex Kim
PhD Student

PS
Priya Shah
PhD Student

ML
Michael Lee
MSc Student

SK
Sara Kim
MSc Student